

Assessment: AI-Assisted Programming

Module

Advanced Programming Concepts

Assignment Title

Analyzing the Role of AI-Assisted Programming in Modern Software Development

Deadline

2026/06/30

Introduction

Artificial Intelligence is increasingly integrated into modern software development through tools such as AI code generators, debugging assistants, and intelligent programming support systems. These technologies are transforming how developers design, implement, test, and maintain software applications.

This assignment is designed to help students understand the impact of AI-assisted programming in software engineering while evaluating both its technical and ethical implications. Students will explore AI-assisted development techniques, implement a programming solution using both traditional and AI-assisted approaches, and critically analyze the results.

Assignment Tasks

Task 1: Conceptual Understanding (Report)

Students must explain:

- What is AI-Assisted Programming?
- Types of AI coding tools (e.g., code generation, debugging, refactoring)
- Advantages in software development
- Risks and limitations (e.g., hallucinated code, dependency, security issues)

Task 2: Practical Implementation (Include screenshots to the Report)

Students are required to:

1. Select a programming problem of moderate complexity
(*Example: Student Management System, Library Management System, Inventory System, etc.*)
2. Develop the solution using TWO approaches:
 - **Version A:** Traditional programming approach (without AI assistance)
 - **Version B:** AI-assisted programming approach using tools such as ChatGPT, GitHub Copilot, or similar AI tools
3. Compare both implementations based on:
 - Development time
 - Code quality
 - Error handling
 - Readability and maintainability
4. Include relevant UML diagrams to improve understanding of the selected problem and system design. Required UML Diagrams
 - Use Case Diagram
 - Class Diagram
 - Activity Diagram
 - Sequence Diagram (*optional for extra clarity*)
5. Students must clearly identify:
 - Which code sections were manually written
 - Which code sections were generated or assisted by AI tools

Task 3: Critical Analysis

Students must discuss:

- How AI changed their coding approach
- Whether AI improved or reduced their understanding
- In which cases AI should NOT be used
- Ethical considerations in AI-generated code

Task 4: Conclusion

- Final opinion on AI in programming education and industry
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Submission Requirements

Students must submit:

- Final report in PDF format
 - UML diagrams (Insert to the report under task 02)
 - Screenshots if necessary
 - AI prompts used during development (Insert to the report under task 02)
 - GitHub repository submission containing: (Insert Link of the repository end of the report)
 - Source code
 - Documentation
 - Project files
 - Commit history
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Formatting & Documentation Instructions

Students must follow the formatting guidelines below when preparing the assignment report.

Element	Recommended Font	Font Size	Style
Main Title	Times New Roman	20 pt	Bold, Centered
Heading 1	Times New Roman	16 pt	Bold

Heading 2	Times New Roman	14 pt	Bold
Paragraph Text	Times New Roman	12 pt	Regular
Code Sections	Consolas / Courier New	11 pt	Regular
Table Content	Times New Roman	11 pt	Regular
Line Spacing	—	1.5	Justified Alignment

Additional Instructions

- Use A4 page size
- Maintain 1-inch page margins on all sides
- Use justified alignment for paragraphs
- Add page numbers
- Include proper headings and subheadings
- Insert screenshots, tables, and diagrams where necessary
- All diagrams must be clearly labeled